RAJADHANI INSTITUTE OF ENGINEERING AND TECHNOLOGY
AUTONOMOUS
Approved by AICTE & Affiliated to APJ Abdul Kalam Technological University
Degree awarded by KTU

KSERC AI Truing-Up Analytical Tool

Industry-Institute Partnership Project | RIET CSE Department

1. What is Truing-Up? (Simple Explanation)

Truing-Up is an annual financial check-up that electricity companies do with KSERC.
Simple Example: You estimate your monthly costs and charge customers accordingly. At year-end, you check actual vs. estimated costs. If you spent less, customers get refunded. If you overspent legitimately, you recover the difference. KSERC verifies this is fair.

Why it matters for Kerala: Truing-up decisions directly affect how much you pay per unit for electricity. A fair truing-up process = fair electricity bills for all consumers.

1.1 Who Are Distribution Licensees? (The Entities Being Audited)

Entities like **Technopark**, **Infopark**, and **Kanan Devan Hills** are not managing KSEB's data. They are **Distribution Licensees**—independent "mini-electricity boards" for their specific areas. Just like KSEB distributes electricity to the general public across Kerala, these organizations are licensed by KSERC to distribute electricity to consumers inside their campuses or estates.

What They Do:

✓ Own and maintain electrical infrastructure (lines, transformers, substations) within their zones
✓ Buy electricity in bulk (usually from KSEB) and retail it to consumers in their area
✓ **Technopark & Infopark:** Manage power supply for IT companies with high reliability
✓ **KDHPCL** (Kanan Devan Hills Plantations Company): Distribute electricity to factories and plantation communities in Munnar

Their Relationship with KSEB:

KSEB	State-wide licensee covering most of Kerala
Other Licensees	Restrict operations to specific boundaries (e.g., Technopark Phase 1 campus)
Connection	They buy high-voltage power from KSEB grid, step it down, and retail to their zones

2. The Problem We're Solving

Before (Manual):	6-12 months | Officers manually review 500+ page documents | 50+ hours data entry | High error risk
After (Our AI):	2-4 weeks | Automated PDF extraction | 10 minutes setup | 100% accuracy with human oversight

1.2 How Truing-Up Works (Financial Process)

Truing-up is a **financial reconciliation** that determines if a licensee (KSEB, Technopark, Infopark) is actually making profit or loss compared to projections.

Step 1	**Projections (ARR & ERC):** KSERC sets theoretical per-unit cost based on licensee's projected expenses for power purchase, maintenance, salaries
Step 2	**Actuals:** Licensee operates for 1 year. Real costs may differ (higher power prices, distribution losses, etc.)
Step 3	**Truing Up:** Licensee submits audited accounts. KSERC checks if expenses were prudent & necessary
Step 4	**Revenue Gap:** If loss found, it's carried forward to next tariff revision for recovery

Key Terms:
• **ARR** = Annual Revenue Requirement (what licensee needs to recover costs)
• **ERC** = Expected Revenue from Charges (estimated income from tariffs)

• **Prudence Check** = Verification that expenses were necessary & not wasteful

3. What We've Built

Live Tool: kserc-trueup-engine-464030854609.asia-south1.run.app
Open Source Code: github.com/prak05/kserc-truing-up

✓ PDF Extraction | ✓ Data Validation | ✓ Prudence Checks | ✓ Report Generation | ✓ Human Override

4. Official KSERC Work Order

Work Order No.	KSERC/CSO/03-05
Date	28 January 2026
Timeline	8 weeks (ISR - No Cost)
Scope	KDHPCL, Technopark, Infopark, CSEZA
Nature	Institutional Social Responsibility (ISR) - Zero Financial Remuneration

Why ISR? RIET is contributing pro-bono to Kerala's regulatory excellence.

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Student-Led Innovation Under Industry Partnership

5. How RIET Students Are Involved

This is NOT just an internship—it's a real project with real impact. RIET students are leading development under the official KSERC Work Order, learning AI, regulatory frameworks, and software engineering simultaneously.

6. Project Leadership

Prakhar Sharma S6 CSE	Full-stack development, AI integration, Claude API, backend architecture. LinkedIn: linkedin.com/in/prak05-btech/
Adwaith A Kumar S4 CSE	Frontend development, UI/UX, data visualization. LinkedIn: linkedin.com/in/adwaithakumar/
Asst. Prof. Akshayraj MR Faculty Mentor	Technical guidance, KSERC liaison, oversight. LinkedIn: linkedin.com/in/akshay-raj-mr/

7. Technology Stack

Frontend: Next.js + React | **AI:** Claude API | **Database:** Supabase | **PDF:** PDF.js + pdfplumber | **Deployment:** Vercel + GitHub | **Fallback:** Gemini, Groq, GitHub Models

8. Why This Matters for NBA Accreditation

Criterion 1: Official work order from KSERC (statutory body)
Criterion 2: Real-world problem solving in AI, regulatory compliance, software engineering
Criterion 5: Pro-bono project serving public welfare through fair tariff-setting
Criterion 7: Potential for academic publications on AI in governance

9. Current Status & Next Steps

✓ **Completed**	MVP live | 4 licensees integrated | KSERC feedback incorporated
■ **In Progress**	FY 2024-25 data processing | Regulatory rule refinements
→ **Next**	Final validation | KSERC publication | Student research papers

Built by RIET for KSERC | Institutional Social Responsibility | Work Order KSERC/CSO/03-05
Tool: kserc-trueup-engine-464030854609.asia-south1.run.app | Code: github.com/prak05/kserc-truing-up